PROGRAM:

server.js

```
const express = require("express");
const http = require("http");
const { Server } = require("socket.io");
const { createClient } = require("redis");
const cors = require("cors");

require("dotenv").config();

const notificationRoutes = require("./routes/notificationRoutes");

const app = express();

const server = http.createServer(app);

const io = new Server(server, {
    cors: {
        origin: "*"
    }
});

app.use(cors());
app.use(express.json());
app.use(express.static("public"));

const redisClient = createClient({
    url: process.env.REDIS_URL
});

redisClient.on("error", (err) => {
    console.log("Redis Error:", err);
});

async function startServer() {

    await redisClient.connect();

    console.log("Redis connected successfully");

    // Socket.IO connection
    io.on("connection", (socket) => {

        console.log("User connected:", socket.id);

        socket.on("registerUser", (userId) => {

            socket.join(`user_${userId}`);

            console.log(`User ${userId} registered`);
        });

        socket.on("disconnect", () => {

            console.log(
                "User disconnected:",
                socket.id
            );
```

```
    });

  });

  // Notification API
  app.use(
    "/api/notifications",
    notificationRoutes(redisClient, io)
  );

  server.listen(process.env.PORT, () => {

    console.log(
      `Server running at http://localhost:${process.env.PORT}`
    );

  });
}

startServer();
```

app.js

```
const socket = io();
const userId = "101";
socket.emit("registerUser", userId);
socket.on("notification", (notification) => {
  const div = document.createElement("div");
  div.innerHTML = `
    <p>
      <strong>${notification.type}</strong>:
      ${notification.message}
    </p>
  `;
  document
    .getElementById("notifications")
    .prepend(div);
});
```

notificationRoutes.js

```
const express = require("express");
const {
  createNotification
} = require("../services/notificationService");
function notificationRoutes(redisClient, io) {
  const router = express.Router();
  router.post("/", async (req, res) => {
    try {
      const { userId, message, type } = req.body;
      const notification = await createNotification(
        redisClient,
        io,
        {
          userId,
          message,
          type
        }
      );
```

```
            res.json({
                success: true,
                notification
            });
        } catch (error) {
            res.status(500).json({
                success: false,
                error: error.message
            });
        }
    });
    return router;
}
module.exports = notificationRoutes;
```

notificationService.js

```
async function createNotification(redisClient, io, notification) {
    const { userId, message, type } = notification;
    const newNotification = {
        userId,
        message,
        type,
        timestamp: new Date().toISOString()
    };
   await redisClient.lPush(
        `notifications:${userId}`,
        JSON.stringify(newNotification)
    );
    io.to(`user_${userId}`).emit(
        "notification",
        newNotification
    );
    return newNotification;
}
module.exports = {
    createNotification
};
```

index.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Redis Notification System</title>
</head>
<body>
    <h1>Real-Time Notifications</h1>
    <div id="notifications"></div>
    <script src="/socket.io/socket.io.js"></script>
    <script src="/app.js"></script>

</body>

</html>
```